



Collective Risk-Sharing: Exploring Real Time Signal Sharing Frameworks in Tandem with a Global Fraud Regulation Fund (GFRF)

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Executive Summary

In the fraud ecosystem, fraud reimbursement liability falls mainly on the Payment Service Provider (PSP), leaving other actors in the “fraud chain” — social media platforms, payment processors, telcos — with no incentive to share financial responsibility or data that could prevent fraud. We propose a polycentric threat detection framework in three points:

- **Implementation of a real-time general aggregate signal-sharing framework** for participants to exchange privacy-preserving *signals* of fraudulent activity, and Personally Identifiable Information (PII) under the GDPR.
- **Establishment of a Global Fraud Regulation Fund (GFRF)**, modelled after bug-bounty-style payouts, to co-ordinate payments and public security certifications to partners that work proactively to prevent fraud.
- **Introduction of collective risk-sharing mechanisms** across fraud ecosystem participants to support proactive fraud disruption and cross-border intelligence coordination.

Background

Financial and governmental institutions increasingly recognise fraud as a collective global issue requiring a coordinated response. For example, PayPal has emphasised the need for a **fraud-preventing data exchange** (PayPal 2024), while Barclays has advocated for regulatory and industry action **aligned under a single overarching framework** (Sneade 2022). The UN’s 2026 Global Fraud Summit reinforced the need **for cross-sector partnerships to distribute risk** and maximise the mutual benefits of collaboration (Skidmore 2026).

Current regulatory frameworks are insufficient. In the United Kingdom and the EU, if you lose your money to a scammer, the bank used in the scam is generally responsible for making you whole. This creates a perverse set of incentives – social media sites and other venues for fraud have no reason to prevent it other than to generally improve the user experience. Each actor in the “fraud chain” has access to different information, but there is no financial or regulatory incentive to share it. This “information cloistering” necessitates a solution that A: enables aggregate information, or “signals” of fraud, to be shared without violating privacy laws, and B: a regulatory framework that incentivizes each actor to share that information.

Proposal

A polycentric threat detection model is proposed wherein individual actors (governments, private firms, systems like GSE and GIRAF, cybersecurity analysts and fraud researchers) proactively share not data per se, but only whether new PII data *matches* data in their system that has previously been associated with fraudulent activity. In PayPal’s case, at the moment of registration of an account, they would receive a report noting whether the phone

number, email address, or other PII used to register the account existed in other actors' fraud-prevention datasets. The reliability of each source's data would be graded with an individual score according to a trustworthiness algorithm trained on a fraud-prevention model, and PII would be shared only when "necessary and sufficient".

To incentivize each individual information cloister to opt in to this system, we recommend legislation that proportionally distributes the liability to refund fraudulent payments across the chain and imposes new fines equal to the cost of that refunded payment. Every firm in the fraud chain will be partially responsible to pay back defrauded individuals, as administered by the PSPs. To borrow from the cybersecurity concept of "bug bounties", these fines will be reserved to pay out to the firms in the chain that successfully stop fraud before it occurs – based on information shared with law enforcement. These successful entities will be rewarded with a publicly accessible "security certification". This will incentivize not just reactive responses to fraud, but proactive prevention of scams.

Analysis

This policy brief answers the question, "is this a better system than what already exists?". As proposed, it succeeds in ways that earlier efforts do not. The key technical achievement of this system is the "match-only" framework, which does not require the actual sharing of data, just a signal of whether the data exists and is considered under a fraudulent matrix as a risk. This elides the critiques of both regulators and critics of digital surveillance: no PII whatsoever is shared between partners.

Creating this framework is a particularly intractable issue because sharing data between industries/agencies/nations is legally tricky. Any firm that does business in the EU or the UK has to deal with the EU's General Data Protection Regulation (GDPR), which is generally considered to be the world's most restrictive (Hoofnagle et al. 2019) and determines how organizations must deal with PII. The GDPR and similar laws in other countries (in our case, Germany, the U.K. and the U.S.) are the reason why other public or private actors in the fraud ecosystem – like Meta, where many victims first meet the scammers - cannot simply give the identities of suspected fraudsters to PayPal, so that PayPal can monitor their activity or prevent them from using their services. The financial incentives for PayPal are strong – per their own SEC filings, they lose perhaps billions of dollars a year to "transaction costs", and these numbers are increasing (SEC, 2023).

The GDPR does have some helpful carve-outs for the sharing of PII. Data can always be shared with law enforcement (ICO 2025). But law enforcement can be slow, especially in an era where the majority of fraud is perpetrated with brand-new AI technologies (PayPal 2026). Aggregate data, and signals of data are also legal to share under the GDPR (Podda 2021).

This allows our system to work significantly faster and more efficiently than the request-based GSE model, which requires laborious regulatory compliance at every step. The key *political* achievement is the establishment of collective risk-sharing systems across all the enablers of fraud, in addition to the GFRF. This "carrot and stick" model incentivizes proactive anti-fraud behaviour from every actor in the chain, rather than the average reactive response. We believe that firms will sign on due to this incentive structure as well as a genuine desire to prevent fraud - leading firms in the "fraud chain" like Meta have already demonstrated an appetite for proactively preventing fraud, as shown by their membership in the GSE.

There are trade-offs, however. Joining the Real Time Signal-Sharing Framework will require some infrastructural build-up and cost for each actor, and there may be initial resistance if firms do not believe that the measure adds any value to their current risk detection models. Likewise, joining a global fund might not be attractive in the long run if actors face greater losses by joining the fund rather than paying reimbursement from their own accounts.



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Appendix

Active link used for the technical live demo: <https://quorum-hq.vercel.app/dashboard>